

RAG Tasks

Task 1: PDF Chat Application

Build app where user uploads PDF and asks questions.

Features

- Upload PDF
- Extract text
- Split into chunks
- Store chunks in MongoDB
- Ask questions
- Return relevant answers

Stack

- Frontend: React
- Backend: Node.js + Express
- DB: MongoDB
- AI: OpenAI / Gemini API

Practice Topics

- File upload using multer
- PDF parsing
- REST APIs
- Embeddings basics

Task 2: FAQ Chatbot

Create chatbot for company FAQs.

Example:

- What is refund policy?
- How to contact support?

Tasks

- Store FAQ docs
- Search matching answer
- Generate natural response

Bonus:

Add admin panel to manage FAQs.

Task 3: Notes Search Assistant

User saves notes and asks:

“Show notes related to AWS”

Features

- CRUD notes
 - Semantic search
 - AI summary
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Task 4: Resume Analyzer

Upload resume and ask:

- What skills are missing?
- Is candidate fit for MERN role?

Features

- Resume upload
- Skill extraction
- JD matching
- AI scoring

Database Collections

users

resumes

job_descriptions

analysis_reports

Task 5: Student Learning Assistant (for tnpLab)

Students upload:

- notes
- PDFs
- assignments

Then ask:

- Explain OS scheduling
- Give practice questions

Advanced Features

- Role based login
- Admin uploads content
- Students ask questions

Great MERN Practice

Includes:

- Auth
- Role management
- AI integration
- Analytics dashboard

Task 6: YouTube Video RAG

Input YouTube URL.

System:

- Fetch transcript
- Chunk transcript
- Store embeddings
- Ask questions

Example:

Summarize this lecture

Useful for training platforms.

Task 7: Legal Document Assistant

Upload:

- contracts
- agreements
- policies

Ask:

- Highlight penalties

- Find payment clauses

Features:

- clause extraction
- citation
- answer grounding

Task 8: Multi-Document Research Assistant

Upload:

- PDFs
- DOCX
- TXT
- CSV

Ask:

Compare revenue growth

Features

- Multi-source retrieval
- Context ranking
- Source references

Task 9: Medical Knowledge Assistant

Upload medical reports and ask questions.

Features:

- OCR
- lab report analysis
- recommendations

(Use with medical disclaimer)

Level 4 — Production-Level Tasks

Task 10: E-Learning RAG SaaS Platform

This is ideal advanced project.

Modules

Admin
Trainer
Student
Super Admin

Features

- Course management
- PDF/video upload
- AI tutor
- AI quiz generation
- AI interview simulator
- Coding challenge generation

Architecture

Frontend:

- React / Next.js

Backend:

- Node.js microservices

Services:

- Auth Service
- Document Service
- RAG Service
- Billing Service

Database:

- MongoDB
- Redis cache
- Vector DB

Cloud:

- AWS S3
- Docker
- Kubernetes

Level 5 — Expert Tasks

Task 11: SQL Database RAG

Natural language to SQL.

Example:

Show top 5 employees by salary

LLM converts to SQL:

```
SELECT * FROM employee ORDER BY salary DESC LIMIT 5;
```

Excellent for MERN + MySQL practice.

Task 12: Codebase RAG

Upload GitHub project.

Ask:

- Explain folder structure
- Find auth flow
- Where is payment code?

Skills

- Code parsing
 - Embedding source code
 - Semantic search
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Task 13: Hybrid Search RAG

Use:

- keyword search
- vector search

Compare results.

Task 14: Agentic RAG

Multiple AI agents:

- Retriever agent
- Evaluator agent
- Answer generator

Great for GenAI.

Task 15: Voice RAG

User speaks question.

Pipeline:

Voice → Text → Retrieval → LLM → Voice Response

Commented [vs1]: Explain